The Great Logan Paleomagnetic Loop — The Polar Wandering Path from Canadian Shield Rocks During the Neohelikian Era[1]

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Normally magnetized dikes and reversely magnetized sills of Neohelikian age near the northwest shore of Lake Superior form two distinct paleomagnetic groups with mean pole positions of 179° W, 35° N, and 140° W, 47° N respectively. Thermal and alternating field demagnetization and the study of magnetic properties and opaque minerals indicate that directions of magnetization of these rocks were acquired at the time of their intrusion. Field evidence suggests that the sills are older than the dikes and radiogenic age determinations indicate that the intrusion of both occurred between 1000 and 1100 m.y. ago.

These pole positions, together with those for the Franklin intrusions pole at 167° E, 08° N (675 m.y.), the Abitibi dikes, at 134° W, 27° N (1150 m.y.), and the Mackenzie igneous events, at 171° W, 4° N (1200 m.y.) are used to define the Logan Loop, the path that the pole took in Neohelikian time relative to the Canadian Shield. Other poles well defined magnetically, but less well dated, from rocks of this era, fit the curve quite well.

Analysis of available data supports the hypothesis that the relative polar movement that gave rise to the Logan Loop was preceded and followed by polar stability vis à vis North America, whereas polar movement may have been quite rapid during the formation of the loop, which coincided with the time of the Grenvillian orogeny. Comparison of the depositional environment of Neohelikian rocks of the Canadian Shield with their probable paleolatitude as indicated by the 5 key points on Logan Loop should provide a test for the validity of this movement.

L'aimantation de dykes et sills néohélikiens (la partie supérieure du Protérozoique Moyen) provenant de la rive nord-occidentale du Lac Supérieur est étudiée. Des dykes à aimantation normale et des sills à aimantation inverse forment deux groupes de paléopôles distincts qui sont respectivement situés à 179° W, 35° N et 140° W, 47° N.

La désaimantation thermique et par champ alternatif, de même que les propriétés magnétiques des minéraux opaques, indiquent que l'aimantation de ces roches fut acquise durant leur intrusion. Bien que sur le terrain, on observe que les dykes sont plus jeunes que les sills, leur âge radiogénique apparent est semblable et oscille entre 1000 et 1100 millions d'années.

Les positions de ces pôles, jointes à celles des intrusions de Franklin (167° E, 08° N, à 675 millions d'années), des dykes de l'Abitibi (134° W, 27° N, pour 1150 millions d'années) et des intrusions du Mackenzie (171° W, 4° N pour 1200 millions d'années) définissent la "Boucle de Logan", le chemin parcouru par le pôle néokélikien pa-rapport au Bouclier Canadien.

D'autres pôles de la région, bien définis magnétiquement, mais moins bien datés, semblent se placer sur la courbe sans difficulté. L'analyse des données indique que les mouvements polaires suivant la "Boucle de Logan" ont été précédés, puis suivis d'une période de stabilité. Par contre, les mouvements polaires, rapides au moment de la "Boucle de Logan", coincident avec l'orogenèse grenvilienne.

La comparaison des paléoenvironnements des roches néohélikiennes du Bouclier Canadien, avec leur paléolatitude probable indiquée par 5 points cruciaux de la "Boucle de Logan", devrait fournir un critère permettant de vérifier la validité de ces mouvements.

Introduction

The pole positions from the Franklin Magnetic Interval (Fahrig *et al.* 1971) at 167° E, 8° N ($\alpha_{95} = 5°$), and the Mackenzie Magnetic Interval (Fahrig and Jones 1969; Robertson 1969), at 171° W, $3\frac{1}{2}$° N, ($\alpha_{95} = 4\frac{1}{2}°$), lie close together. These pole positions were derived from detailed paleomagnetic work on numerous samples taken from a wide area of the Canadian Shield and are considered to be highly reliable. The mean age

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of the Franklin Magnetic Interval is 675 m.y. (Fahrig *et al.* 1971), and that of the Mackenzie Magnetic Interval 1200 m.y. (Fahrig and Jones 1969). In the intervening 525 m.y. the pole relative to the Canadian Shield appears to have moved in a great loop in the North Pacific.

The Logan dikes and Logan sills studied in this paper provide two critical points on the loop. Most of the dikes are magnetized normally, whereas all the sills are reversely magnetized. If polarity is ignored, the normally magnetized dikes and reversely magnetized sills exhibit a difference in pole position of about 30°, which may be explained by one of three possibilities:

(1) The earth's total magnetic field could have an asymmetric, non-dipole, long term component (Wilson 1970). This possibility could be tested by obtaining directions of both polarities

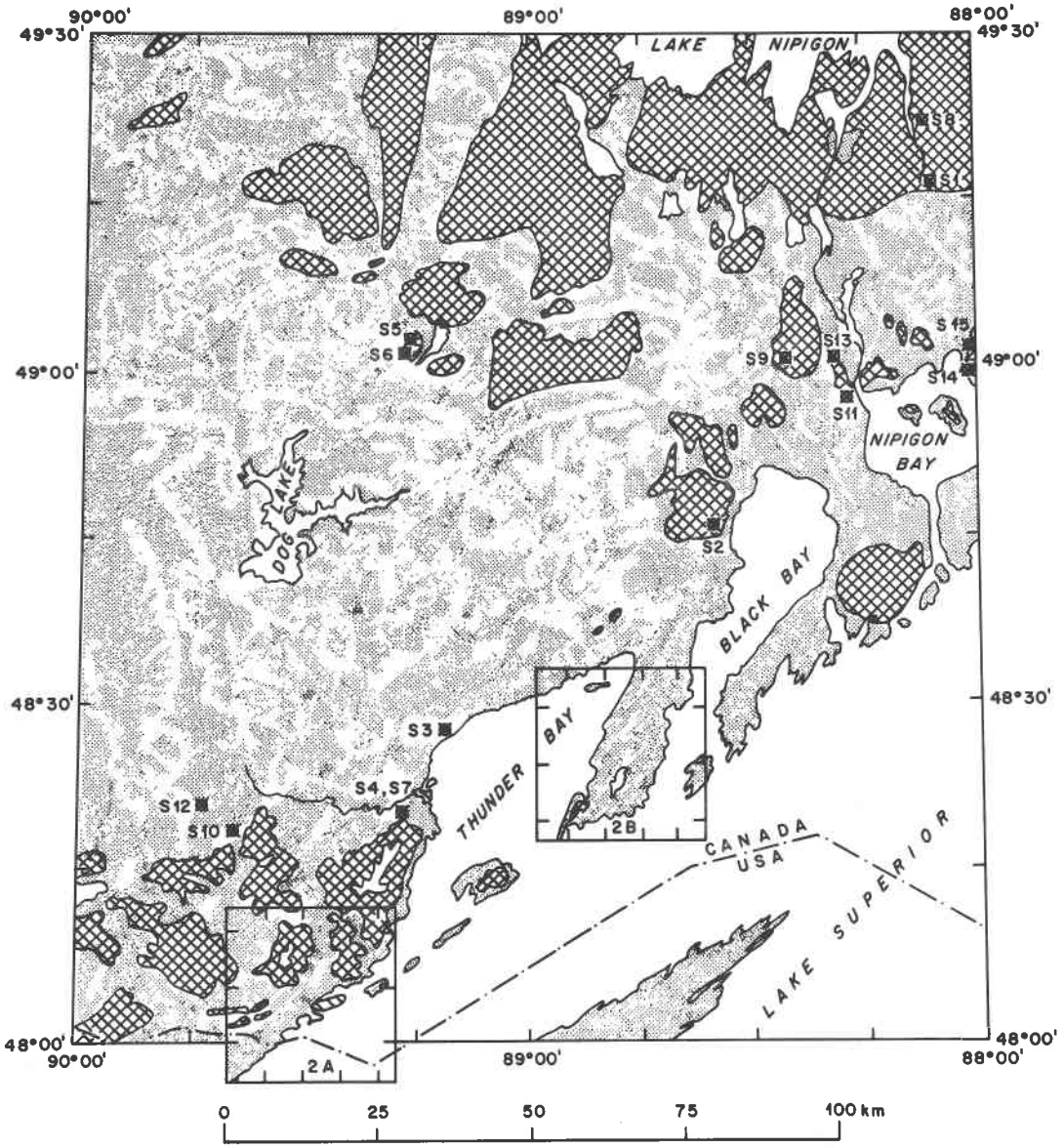


FIG. 1. Locality map showing generalized outcrop of the Legion sills and the sites where they were sampled, also outlines of Figs. 2*a* and *b*.

at two different ages and seeing whether a similar discrepancy between normal and reversed poles occurred at both times. The usual assumption of an axial dipole field ignores this possibility.

(2) The normally and reversely magnetized rocks might both be affected by a secondary component that would add to one and subtract from the other (Palmer 1970). A component of this sort should cause streaking of directions away from it, apparent during laboratory demagnetization. No such streaking was observed during extensive laboratory tests, and the distinct discrete groups, after cleaning (Figs. 5*a* and *b*), make this possibility unlikely.

(3) The difference in pole position between the normally and reversely magnetized rocks could be the result of a relative polar shift between the times at which the two sets of rocks acquired their magnetization. These times are not greatly different; the actual difference cannot at present be accurately determined (an error of 5% in the determined age at this time will give an error of ± 50 m.y.) but is unlikely to be more than 100 m.y. This gives an average minimum rate of movement of the pole between the intrusion of the two sets of rocks of about 0.33 cm/year, but it may have been much more than this.

Geological Setting, Age, Sampling and Measurement

The Logan sills and dikes intrude flat-lying Animikie sediments of Aphebian age. In the Pigeon River area (Fig. 2*a*) the intruded sediments are Rove shales and graywackes, for which an Rb/Sr isochron has yielded an age of 1664 m.y. (Peterman 1966). In the Sibley Peninsula and north of Nipigon (Figs. 1 and 2*b*), the sills and dikes intrude pink limy mudstones of the Sibley Series. R. K. Wanless (personal communication, 1970) has obtained an Rb/Sr isocron from them giving an age of 1298 $\pm$ 33 m.y.

The sills are flat lying, intruding bedding planes of the Sibley Series, but transgressions from one plane to another are seen in places. The K/Ar radiogenic age determinations from them, given in Table 8, range from 730 to 1060 m.y.; however, these are whole-rock analyses, some argon loss is likely, especially from the coarse-grained samples, so these must be considered to be minimum ages. A Rb/Sr age, based on an isochron obtained from rock samples taken from the paleomagnetic sites, confirming the K/Ar age of the sills, will be published shortly by members of Carleton University.

The dikes are essentially vertical, and range in width from 1 m to more than 100 m. The majority, including all the normally magnetized dikes sampled, strike northeast, but a few, including two reversely magnetized dikes, strike northwest. Three K/Ar radiogenic ages on biotites from the normally magnetized dikes give a mean age of 1016 m.y. (Table 8) and two whole-rock determinations give a mean of 1020 m.y. Hence, the age assigned to this swarm is 1020 m.y.

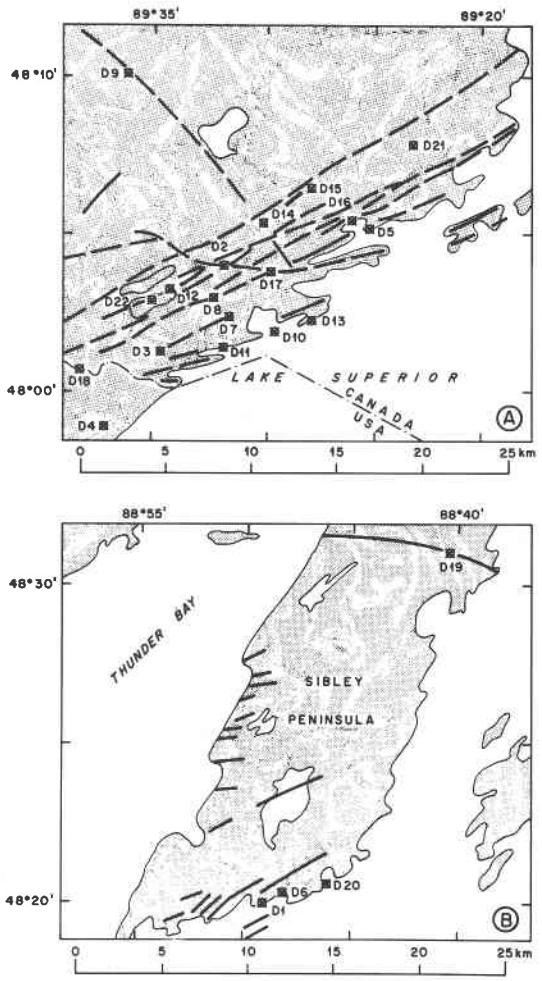


FIG. 2. (*a*) Locality map showing dike sampling sites in the Pigeon River area, and the major dikes. (*b*) Locality map of the Sibley Peninsula showing the major dikes and the sampling sites.

Both dikes and sills are diabase or gabbro, with fine-grained margins, and quite commonly with granophyric segregations. Most samples were collected by coring in the field, using a portable drill (Gaucher and Meilleur 1965), but some of the sill samples were collected by hand. The samples were oriented by suncompass (Haaf and Wensink 1962), checked by a Brunton compass, and the orientation is considered to be correct to better than 2°. Wherever possible 5 or more samples were collected from each site to enable the calculation of precise within-site statistics, but sites with 2 or more concordant samples were accepted.

Some of the measurements were made on the automatic bi-astatic magnetometer of the Geological Survey of Canada (Larochelle and Christie 1967), while others were made on the Princeton Applied Research spinner magnetometer of the Earth Physics Branch, and the most weakly magnetized samples were measured on the astatic magnetometer of the Earth Physics Branch designed by J. L. Roy *et al.* (1971*b*). The three instruments gave results that were in good agreement in both direction and intensity. Alternating field demagnetization was performed using both the Earth Physics Branch demagnetizer (J. L. Roy *et al.* (1971*a*)) and that designed by the paleomagnetic staff of the Geological Survey of Canada (Larochelle and Black 1965). Thermal demagnetization was carried out in the non-magnetic oven designed by J. L. Roy *et al.* (1971*c*).

The Normally Magnetized Dikes

A suite of northeasterly striking dikes outcrops near the coast of Lake Superior near the Pigeon River and on the coast of the Sibley Peninsula (Figs. 2*a* and *b*). Most of these dikes appear to constitute a single phase of activity characterized by positive magnetization directions with west–northwesterly declination and intermediate inclination. Oriented samples were collected from 20 sites from at least 15 dikes. Fifteen of these sites form a coherent group, which constitute the normally magnetized dikes of this paper. At two sites (D3 and D6) the rocks were reversely magnetized and they are considered in a later section. Two sites (D10, D12) gave random directions at $P = 0.95$ (Watson 1956) and were excluded from the calculations. Site D20 was excluded because it seemed likely that the anomalous direction resulted from errors in the declination caused by deviations of the magnetic compass, which had to be used as fog precluded the use of a suncompass.

Thermal Demagnetization

One specimen from each of the 7 sites was heated in incremental steps and the magnetization measured after each step. The normalized demagnetization curves given in Fig. 3*b* show in most cases a discrete component with a blocking temperature above 500° C and a component distributed between 50° and 550° C (c.f. Roy *et al.* 1967). Distributed components dominate in sites D5 and D10, whereas discrete components are prominent in sites D1, D2, D4, and D8. Direction changes throughout the range of temperature of 20° to 550° C are very small, and the mean for the group remains close to 296,+43; this suggests that the discrete and distributed components are magnetized in the same direction and that secondary components are unimportant. The discrete component with a blocking temperature above 500° C is attributed to nearly pure magnetite, which acquired its magnetization as a TRM (thermoremanent magnetization), or high-temperature CRM (chemical remanent magnetization) (Carmichael and Nicholls 1967) during the initial cooling of the dikes.

Opaque Mineralogy

Details of examination of polished sections from 7 sites of this group are shown in Table 5. The oxides are large, ranging up to 1000 microns in most sections, and typically anhedral, although skeletal grains are common in sections from sites D1 and D10. Ilmenite is prominant, and magnetite less so in all sections. Magnetite–ilmenite intergrowths are commonly present, but the amount varies widely from site to site. The oxides at sites D1 and D7 are in a low oxidation state, whereas at other sites intermediate oxidation is characteristic. Maghemite is either absent, or at sites D7 and D8, present in very small amount. Small sulfide crystals, of which pyrrhotite is predominant, appear to be a universal feature of these dikes. In this small sample there appears to be no correlation between oxidation state and magnetic stability, but sites with low oxidation state appear to contain the largest proportion of distributed component.

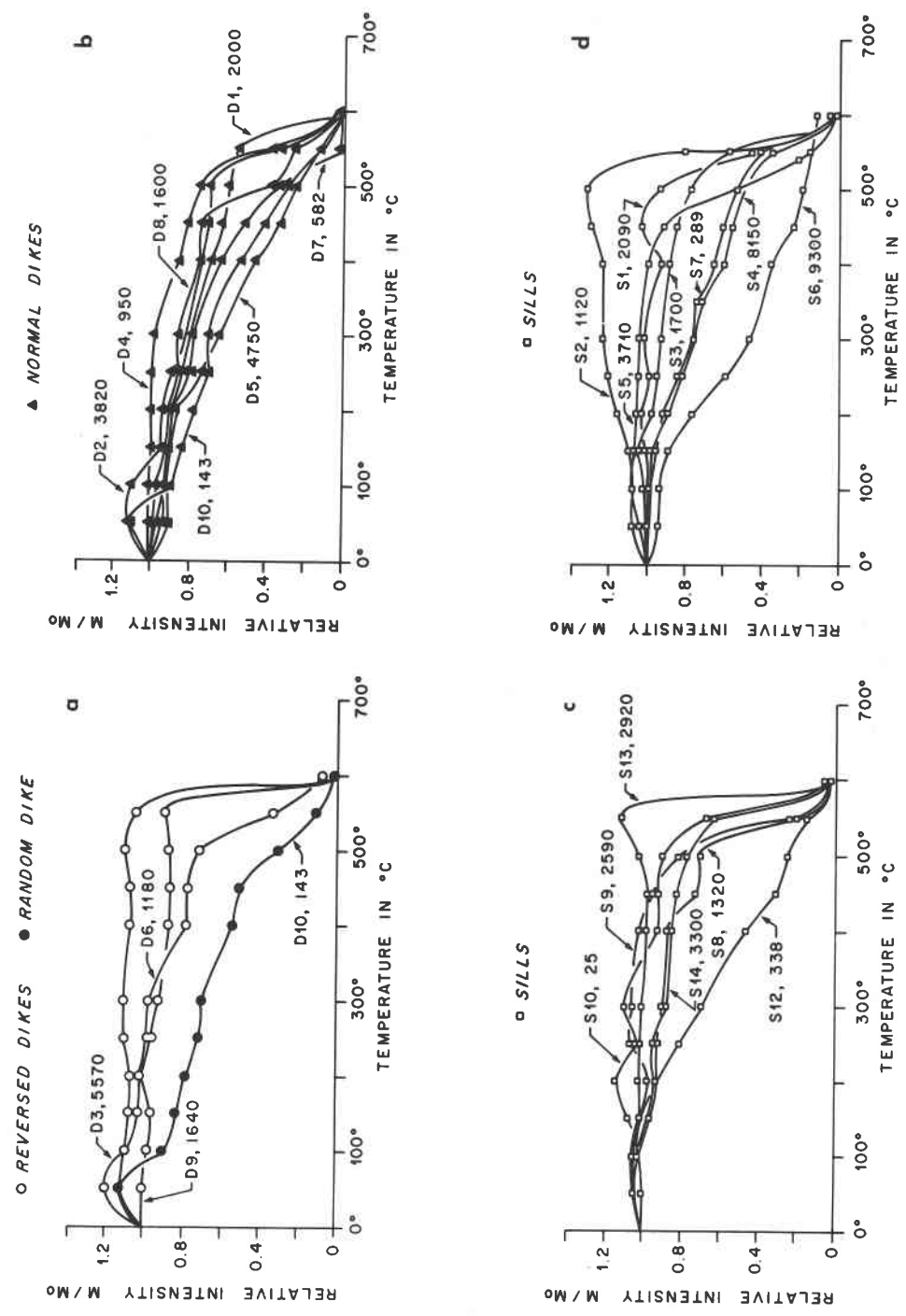


FIG. 3. Thermal demagnetization curves of a specimen from each site. The number beside the site number is the NRM intensity in e.m.u. $\times 10^{-6}$ for the specimen.

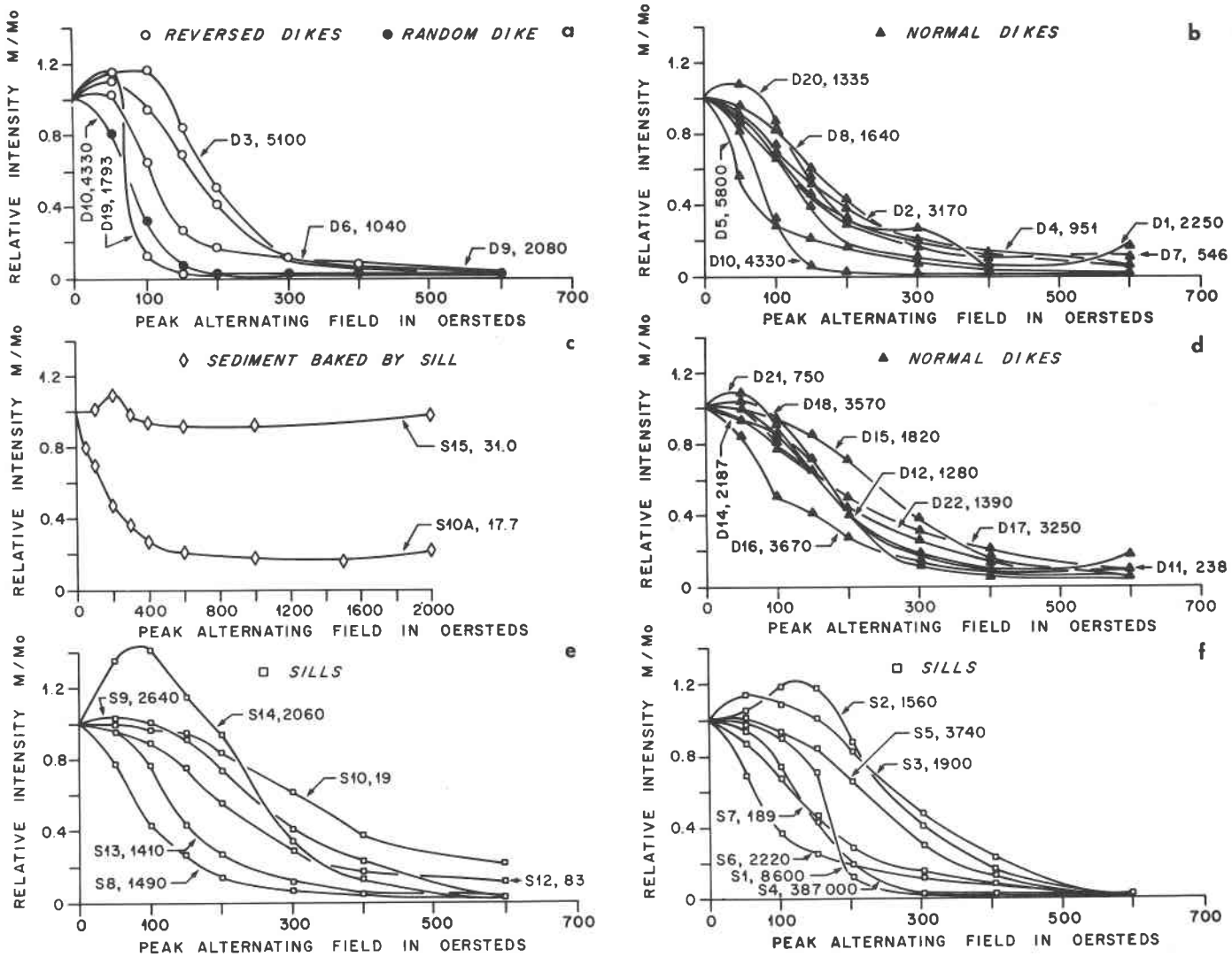


FIG. 4. Alternating field demagnetization curves of a specimen from each site. The number beside the site number is the NRM intensity in e.m.u. $\times 10^{-6}$ for the specimen.

TABLE 1. Normal dike directions

Site	N	M	D°	I°	k	α	A.F.	N	M	D°	I°	k	α
D1	5	1151	293	+48	13	22	150	5	424	287	+36	31	14
D2	5	3666	295	+47	138	6.5	150	5	1305	300	+36	457	3.6
D4	6	1814	292	+49	104	6.6	150	6	871	288	+43	89	7.1
D5	5	3134	291	+14	2.1	72	150	5	1044	287	+47	358	4.0
D7	4	853	308	+46	50	13	150	4	387	300	+38	381	4.7
D8	5	1151	312	+54	48	11	150	5	596	303	+46	352	4.1
D11	6	87	296	+45	151	5.5	300	6	15	294	+43	31	12
D13	3	1.6	302	+58	285	7.3	200	3	1.5	305	+60	35	21
D15	6	1698	292	+56	14	18	300	6	577	306	+50	229	4.4
D16	5	20090	255	+64	4.3	42	300	5	1205	287	+61	28	15
D17	4	1108	285	+26	1.5	145	300	4	320	288	+38	19	22
D18	5	3447	307	+47	233	5.1	300	5	842	306	+46	190	5.6
D21	3	552	278	+57	6.0	56	300	3	111	300	+46	16	32
D14	4	109200	285	+37	8.2	34	300	4	1384	289	+32	25	19
D22	3	1013	301	+40	87	13	300	3	237	293	+24	46	18
Sites	15	9930	293	+47	28.8	7.2	—	15	621	295	+43	53.6	5.3
Samples	69	9190	294	+48	8.3	6.3	—	69	656	295	+44	31.9	3.1
Excluded dikes													
D10	5	973	13	+37	1.1	—	400	5	4.4	229	+45	1.4	115
D12	4	54840	160	+79	0.9	—	300	4	23000	273	+72	2.2	84
D20	5	870	317	+21	14	22	200	5	294	311	+ 7	17	19

N = the number of samples measured.
M = the arithmetic mean intensity of magnetization of the samples in e.m.u. $\times 10^{-6}$.
D° = the declination clockwise from true north.
I° = the inclination positive downwards.
k = Fisher's (1953) best estimate of precision.
α = the half angle of the cone of confidence at $P = 0.95$ (Fisher 1953).
A.F. = the peak alternating magnetic field in oersteds.

Alternating Field Magnetization

One pilot specimen from each of the tightly grouped sites, and two from those whose initial directions were scattered, were demagnetized in an alternating magnetic field increased in steps, in the absence of an ambient field. Analysis of the direction changes in these pilot runs determined the peak alternating field used for the magnetic cleaning of each site.

A normalized demagnetization curve for a specimen from each site is given in Figs. 4*b* and *d*. With the exception of sites D1, D5, D10, and D16, the coercivity spectra are quite high. Site mean directions and precisions before and after magnetic cleaning are given in Table 1, together with the group means. In general, the within site precision was greatly increased by the magnetic cleaning, but the mean direction was not significantly changed. This means that a random component was removed by the cleaning; it seems likely that the random component was predominantly a temporary component acquired in the earth's field during storage in the

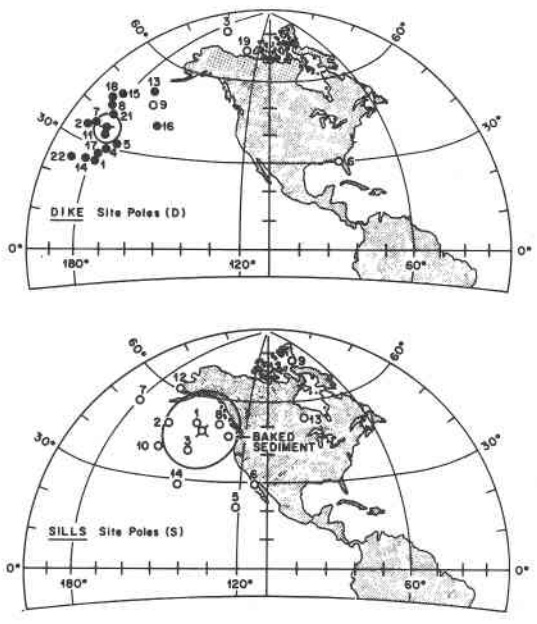


FIG. 5. An equal area projection of the western hemisphere (20° W–160° E) showing site pole positions for (*a*) dikes and (*b*) sills. The mean pole with the P = 0.95 error oval is also shown for the two major groups.

TABLE 2. Magnetic properties

Site	n.r.m. $\times 10^{-3}$	m.d.f. n.r.m.	a.r.m. $\times 10^{-3}$	m.d.f. a.r.m.	i.r.m. $\times 10^{-3}$	m.d.f. i.r.m.	Fe_2O_3	FeO	S.G.
D1	2.25	130	13.7	155	455	270	1.89	11.56	3.00
D2	3.17	155	9.6	60	1010	280	1.96	10.98	3.02
D4	0.95	135	8.8	120	188	260	2.05	9.59	2.96
D5	5.80	55	17.0	195	620	330	1.63	10.29	3.03
D7	0.55	160	7.0	180	265	320	—	—	3.00
D8	1.64	175	12.7	175	485	320	3.1	10.7	3.01
D3	5.10	200	26.8	160	205	170	4.73	11.25	2.99
D6	1.04	180	23.3	220	1036	350	—	—	3.03
D9	2.08	135	13.9	135	400	220	1.6	10.1	3.01
D10	4.33	70	1.0	120	27	260	—	—	2.79
S1	1.60	—	17.3	140	510	320	2.07	9.76	3.03
S2	1.56	275	23.3	200	940	330	2.91	9.66	3.02
S3	1.90	290	11.2	190	485	405	2.19	7.77	2.94
S4	387.0	130	27.0	195	1150	345	3.84	11.40	2.95
S5	3.74	240	22.4	185	835	350	2.55	10.78	3.03
S6	2.22	70	24.7	140	910	310	2.79	11.69	3.03
S7	0.07	70	2.3	240	35	320	1.8	13.6	2.94
S8	1.49	90	10.7	105	290	205	2.10	8.16	3.01
S9	2.64	270	16.5	245	695	390	2.05	9.85	3.02
S10	0.02	350	0.3	435	5	620	0.70	4.73	2.86
S11	73.6	90	22.5	185	890	330	2.52	10.40	3.07
S12	0.08	210	2.4	315	21	335	2.41	12.73	2.96
S13	1.41	140	21.2	185	868	360	2.71	10.10	3.02
S14	2.06	265	45.2	175	1634	300	4.39	6.94	2.90
Baked sediment	0.01	280	0.7	850	8	1000	0.70	4.73	—

The a.r.m. is the anhysteritic remanent magnetization acquired in an alternating field of 2500 Oe and a bias steady field of 0.3 Oe.
The n.r.m. is the natural remanent magnetization in electromagnetic units.
The m.d.f. is the median destructive field in oersteds.
The i.r.m. is the isothermal remanent magnetization induced in 11 000 oersteds.
The S.G. is the specific gravity.
In the case of n.r.m. and a.r.m. the m.d.f. is the field in oersteds required to reduce the initial intensity by half. In the case of i.r.m. the m.d.f. is the field in oersteds required to reduce the mean intensity to zero; in this case the lower coercivity grains are magnetized oppositely to the higher coercivity grains.

laboratory. Site pole positions, calculated from the mean cleaned site directions of Table 1, are plotted in Fig. 5*a*.

General Considerations

Some magnetic properties are given in Table 2. Specimens with natural remanent magnetization / anhysteritic remanent magnetization (NRM / ARM) ratios near unity are considered to be rocks with an uncomplicated magnetic history (i.e. D1, D4, D7, D8); where the NRM/ARM ratio is low, magnetic stability is likely to be low, as in sites D5 and D10, but as the cleaned results from site 5 show (Table 1), this does not necessarily mean that a valid direction cannot be obtained. The variation of specific gravity between the dikes and the sills is too small to be correlated with magnetic characteristics. The ferrous/ferric ratio is quite high for the rocks as a whole, indicating that the rocks are in quite a reduced state and bears no close relationship to the oxidation state of the opaque minerals.

The high blocking temperatures (Fig. 3*b*) and coercivity spectra (Figs. 4*b* and *d*) for most of the sites, together with simple opaque mineralogy, showing no appreciable chemical alteration, combined with good within and between site groupings from dikes cropping out over a considerable area (Figs. 2*a* and *b*), add up to a very high probability that this group accurately reflects the mean field (the dipole field) at the time of intrusion. We therefore consider that the pole for this group of dikes given in Fig. 5 and the 95% error oval associated with it, are realistic estimates of the pole relative to this area of the Canadian Shield at the time the dikes cooled,

TABLE 3. Sill thermal demagnetization

Temp.	M	D°	I°	k	α
NRM	1900	98	−59	9.5	16.5
50° C	1990	101	−59	8.9	17.1
100° C	1990	105	−57	8.4	17.7
200° C	1940	103	−62	11.5	14.9
300° C	1900	104	−62	14.2	13.3
400° C	1820	107	−63	15.2	12.8
450° C	1780	103	−61	19.1	11.3
500° C	1330	102	−64	18.2	11.6
550° C	1060	101	−63	17.4	11.9
600° C	29	92	−22	1.9	48.7

These are means of 1 sample from each of the 10 sites S1, S2, S3, S5, S7, S8, S9, S10, S13, and S14.
Temp. is the temperature to which the specimens were raised; they were cooled in zero field and kept in magnetic shield containers until measurement.
M = the arithmetic mean intensity of magnetization in e.m.u. $\times 10^{-6}$.
D° = the mean declination measured clockwise from true north.
I° = the mean inclination, negative upwards.
k = Fisher's best estimate of precision.
α = the half angle of the cone of confidence at $P = 0.95$ (Fisher 1953).

that this is significantly different from that for the younger Franklin event, and from that of the intrusion of the sills, a slightly older event.

The Logan Sills

Dubois (1962) and more recently Palmer (1970) obtained a pole position for these sills, which is near Queen Charlotte Islands in the northeast Pacific. This pole was from samples that were reversely magnetized relative to the present field and to the dikes of the previous section. However, this is not an exact reversal relative to the dikes as the pole positions for the two groups (Fig. 5) (ignoring polarity) differ by 31°. The initial directions from these sills were widely scattered, and Palmer suggested that the difference in pole positions between them and the normally magnetized dikes could be caused by a normal secondary component along a recent field direction affecting both sets, adding to the magnetization of the normal dikes and subtracting from and scattering the magnetization of the reverse sills. If such is the case then the correct pole of the field at the time will be given approximately by the mean of the two poles. On the other hand the sills are shown by crosscutting relationships (Tanton 1931; DuBois 1962) to be older than the normally magnetized dikes, so that the difference in their pole positions may be a genuine difference caused by a relative polar shift between the times the two sets of intrusions acquired their magnetization.

This study attempts to make a detailed analysis of the magnetization and to eliminate the possibility that the sills retain a secondary component, before calculating the final pole position. The sites were spread over a wide area (Fig. 1) and in many cases contacts were seen to be horizontal within 2°, so that any tilting after cooling is likely to be very small. The areal distribution, petrological affinity, and paleomagnetic consistency leave no doubt that these sills (and the reversely magnetized associated dikes) form a single phase of igneous activity.

Thermal Demagnetization

A specimen from each of 13 sites was thermally demagnetized in steps, and the normalized intensity decay curves are shown in Figs. 3*c* and *d*. An initial increase in intensity at most sites after the 50° and 100° C heating is the result of the removal of a small viscous component along the earth's field direction, nearly opposite in direction to the NRM. This component, which would separate the sills pole from the normal dikes pole, is removed in temperatures below 200° C (see Fig. 3) and is too small a proportion of the total appreciably to change the mean direction (Table 3). Most of the sites have a large discrete component with a blocking temperature between 500° and 600° C. Notable exceptions are sites S6 and S12, which have a large distributed component, and also a very wide initial scatter (Table 4). The two sites (S2 and S13) with no distributed component (the increase in intensity is due to the removal of a small viscous component which reduced the NRM value) are

TABLE 4. Reversed directions

Site	N	M	D°	I°	k	α	A.F.	N	M	D°	I°	k	α
Sills													
S1	5	1140	77	− 3	2.4	65	400	5	69	111	−71	71	9
S8	4	1690	145	−26	2.9	66	400	4	66	108	−75	77	11
S2	5	1440	124	−65	12.0	23	400	5	235	116	−64	90	8
S5	7	3300	12	−72	12.5	18	400	7	417	53	−69	70	7
S6	5	11200	294	−74	2.5	62	400	5	224	53	−75	65	10
S9	2	2450	192	−75	9.4	94	400	2	556	183	−76	107	24
S13	5	1390	40	−70	1.2	—	400	5	144	148	−88	10	26
S14	4	3620	79	−70	10.7	30	400	4	264	84	−62	47	14
S3	6	713	108	−80	2.5	53	400	6	160	97	−68	71	8
S7	4	99	128	−53	66	11	400	4	7	131	−55	92	10
S10	2	21	110	−55	1400	7	400	2	10	105	−62	558	11
S12	2	144	262	−55	3.2	—	400	2	12	134	−66	17	66
Sites	12	2270	109	−73	5.5	20	400	12	180	107	−72	34	8
Samples	51	—	103	−72	2.9	14	400	51	—	100	−72	23	4.3
Sediment baked by sill	2	19.8	112	−42	2.2	—	2000	2	16.1	99	−76	90	27
Sill sites excluded													
S4	5	104 900	131	− 5	1.7	87	400	5	840	111	−48	2.4	63
S11	1	Only one sample, hence site statistics impossible											
Dikes													
D3	6	4620	159	−67	76	8	400	6	275	161	−67	69	8
D6	5	1110	293	− 3	39	12	400	5	98	348	−76	108	7
D9	6	7030	126	−63	2.4	55	400	6	136	117	−60	154	5
D19	6	9030	226	−69	2.8	49	400	6	100	156	−74	45	10
Sites	4	5450	234	−49	1.1	—	400	4	152	134	−77	17	23
Samples	23	5640	235	−54	1.3	50	400	23	155	135	−76	18	7.3

N = the number of samples measured.
M = the arithmetic mean intensity of magnetization of the samples in e.m.u. $\times 10^{-6}$.
D° = the declination clockwise from true north.
I° = the inclination positive downwards.
k = Fisher's (1953) best estimate of precision.
α = the half angle of the cone of confidence at $P = 0.95$ (Fisher 1953).
A.F. = the peak alternating magnetic field in oersteds.

ones in which the characteristic oxide is a fine magnetite–ilmenite lamellar intergrowth (Table 5), whereas those with a large distributed component (with the exception of S6 which has an anomalously high NRM probably due to a nearby lightning strike) S12, S4, and S7, contain oxides in the lowest oxidation state.

The mean direction from the 10 specimens from 10 sites that were thermally demagnetized by steps did not change significantly at any step up to and including 550° C (Table 3). This high thermal stability indicates that either the distributed component is in the same direction as the discrete component, or that it forms an insignificant portion of the magnetic vector. The discrete component may almost certainly be attributed to lamellar intergrowths of magnetite, which acquired their magnetization either as a genuine TRM while cooling, or as a CRM during the late stages of cooling (Carmichael and Nicholls 1967), and in either case the direction would be that of the earth's field at the birth of the sills.

Opaque Mineralogy

Results of examination of one polished section from each site are summarized in Table 5. Site S11 is close to a mineral zone and is anomalous; no sulfides were visible in the polished section, but as only one sample was available this site

TABLE 5. Opaque minerals

Site	Percentage opaques	Size range of oxides in microns	Shape	Oxidation state	Magnetite–ilmenite intergrowths	Magnetite	Ilmenite	Hematite	Ulvöspinel	*Sulfides* Size in Microns	Pyrrho.	Pyrite	Chalco.	Other	Remarks
D1	1.4	10– 60	K	C1	—	0	×	—	—	2– 30	•	—	—	—	Skeletal and graphic intergrowths with transparent mineral.
D2	3.8	50– 800	A	C4–5	LX	0	+	—	—	5– 50	0	•	•	—	
D4	0.8	50–1000	A	C3–4	L+	0	×	—	—	1– 5	•	•	•	—	Some aenigmatite. Small magnetites and pyrrhotites along cracks.
D5	2.0	50–1000	A	C3–4	L+	0	×	—	*	5– 60	0	—	—	—	Some aenigmatite. Opaques graphically intergrown with transparent minerals.
D7	2.2	50–1000	C–A	C1	I0	M0	×	—	—	2– 20	0	•	•	—	Common aenigmatite. Sulfides along cracks.
D8	2.5	50– 800	A	C3	I+L	M+	+	—	*	5– 50	•	•	•	—	Small magnetites along F.M. cracks. Spinel along 111 magnetite planes.
D3	1.2	10– 900	A	C3–4	LX	M0	+	—	—	5– 60	•	•	•	—	
D6	2.8	20– 450	EA	C3–4	LX	M.	0	—	—	20– 50	0	•	•	—	
D9	1.3	100–1000	K–A	C3	LX	•	+	•	—	5– 60	•	—	•	—	Common aenigmatite. Hematite in fine cracks.
D10	2.3	50–1000	K	C2–3	L0	+	+	—	—	10–400	+	•	•	—	
S1	2.8	100– 600	A	C3	L+	•	×	•	*	1–100	•	•	—	—	
S2	0.8	100–1200	EA	C3	LX	M.	•	—	—	5– 20	—	•	—	—	Large aenigmatites. Small magnetites along cracks.
S3	1.0	60– 200	C+E	C3	LX	•	•	—	—	5– 50	•	•	•	—	Myrmekitic inter growthsof ilmenite sphene.
S4	3.0	5– 100	K	C1	—	0	×	—	—	2– 6	•	•	—	—	
S5	1.3	100–1000	EA	C2–3	I+L	+	•	•	—	1– 60	•	•	•	—	Crack fillings of magnetite and hematite.
S6	3.5	50–1300	A	C4	LX	—	•	—	—	5– 50	—	—	•	—	Aenigmatite present.
S7	3.7	10– 100	K	C1	—	0	X	—	—	1– 80	•	—	—	—	Pyrrhotite in clusters.
S8	1.2	200– 900	C+A	C2	IXL	0	+	—	—	10– 60	0	—	—	—	Myrmekitic magnetite-spinel intergrowths.
S9	1.8	100–1000	A	C1–2	IX	+	0	—	—	5– 60	—	•	—	—	
S10	1.8	10– 100	E+A	C1	—	•	×	—	—	6– 40	+	—	—	—	
S11	2.5	60– 300	C+A	C2	IXL	M0	0	—	*		—	—	—	—	
S12	1.2	10– 100	E	C1	—	0	×	—	—	5– 15	0	—	—	—	
S13	3.2	100–1000	C+A	C4	LX	0	0	—	*	5– 20	•	—	—	—	Many irregular shrinkage cracks.
S14	2.6	10– 150	C+K	C2	IX	0	0	—	—	2– 6	•	—	—	—	Opaques wrapped around transparent minerals.
Contact	1.8	10– 80	E	—	—	0	—	—	—	5– 60	×	—	—	—	

Shape E = Euhedral, EA = Subhedral, A = Anhedral, C = Corroded, K = Skeletal.
Oxidation state C1 = Unoxidized, grading up to C6 = Highest oxidation state (see Wilson and Haggerty 1966).
Mineral proportions × = >50%, + = 25–50%, 0 = 5–25%, • = <5%, — = absent, proportions of total opaques in section.
*Presence of ulvöspinel is indicated by wormlike runnels in the magnetite-portion of magnetite–ilmenite intergrowths.
In the magnetite–ilmenite intergrowths L = lamellar intergrowths and I = irregular intergrowths.
M is evidence of some maghematization of the magnetite.

was excluded from the statistics. All the coarse-grained sections contain many corroded or anhedral grains composed of fine magnetite–ilmenite lamellar intergrowths, indicating an intermediate oxidation state (Wilson and Haggerty 1966). The sites exhibiting low oxidation state (S4, S7, S10, and S12) contain fine-grained oxides. All sections contain ilmenite and less commonly magnetite as individual grains. Small sulfide grains are characteristic of this intrusive suite, pyrrhotite being the most common, but invariably subordinate to magnetite to the extent that it could not have a significant effect on the magnetic properties. Corroded oxides and intergrowths of the oxides with spinel and sphene indicate some interaction with the residual liquid during high temperature cooling, but low temperature alteration is virtually absent.

Alternating Field Demagnetization

At least one specimen from each site was demagnetized in zero ambient field in an alternating magnetic field increased in steps up to 600 Oe peak. A surprisingly wide spread of coercivity spectra displayed in Figs. 4*e* and *f* may be accounted for by a variable amount of almost oppositely sensed viscous component, removed in low and intermediate alternating fields, causing a rise of intensity at 50 and 100 Oe for sites S2, S3, S9, and S14.

The high NRM intensities at sites S1, S4, and S6 are probably the consequence of a nearby lightning strike, which also caused a scatter of the initial directions. Possibly the large scatter at sites S8 and S13 were caused by a somewhat more remote lightning strike. The remaining sites retain at least 10% of their magnetization after cleaning in 400 Oe. Table 4 shows the initial site mean directions and dispersions, and those after magnetic cleaning in 400 Oe. In most cases the site dispersions are greatly reduced, and yet the formation mean is not appreciably changed. We may infer from this that the net result of lightning strikes (many of the sites are in exposed places) and a reversely directed viscous component, is the addition of random components, which will average out over a sufficiently large sample and give a valid mean from NRM directions. However, in order to sharpen the statistics so that small differences in pole position may be detected, a cleaning technique such as that adopted here is essential. The resulting site pole positions are given in Fig. 5*b*. Much of the increased scatter of these site poles relative to those of the normal dikes is due to the decrease in precision of definition of the pole that follows from the geometry of sites in which the inclination is high (Irving 1964, p. 70).

Baked Contacts

Two samples of sedimentary rock baked by the overlying sill were obtained at each of sites S10 and S15. The intruded sediment at site S10 is a fissile gray shale of the Rove Formation. The samples were taken from within 15 cm of the lower contact, where the shale has been hardened and spotted. At site S15 the sill has intruded pink marl of the Sibley Series. The samples come from about 20 ft (~ 6 m) below the base of the sill in a zone that has been baked by the sill to a distinctive purple color.

At both sites the coercivity is extremely high (Fig. 4*c*). At site S15 no appreciable decrease in intensity was obtained nor change in magnetization direction, after alternating field cleaning in 2000 Oe peak. Examination of a polished section from this site showed a strong red color indicating a high oxidation state, and presumably caused by extremely fine hematite. The only opaque minerals identifiable in the section were anhedral fretted hematite grains in the size range 1–15 microns; these appeared to be somewhat larger and more irregular than those in unbaked Sibley sediments. The magnetization at site S10 resides mainly in fine-grained pyrrhotite. Initially scattered directions rotate towards (107–70), after alternating field cleaning in 2000 Oe, which is close to the mean for the sills.

While there is no evidence that these directions are significantly different from those of the Rove Formation, direction unknown, and the Sibley Series, direction uncertain (although rocks of similar age elsewhere, such as K, L, and M of Table 7, have quite different directions), they have both visibly been baked by the sill, which almost certainly gave them their magnetization. That the direction of this magnetization coincides with that of the sill is considered very strong evidence that the direction from the sills is primary, acquired at the time of cooling and baking.

The Reversely Magnetized Dikes

Four dikes (D3, D6, D9, and D19) yielded reversely magnetized directions, similar to that

TABLE 6. Anchor pole positions

Symbol	Formation	Polarity	Assigned age (m.y.)	S	Pole position	A_{95}	References
1	Franklin	M	675	30	167° E, 8° N	5	Fahrig *et al.* (1971)
2	Normal dikes	N	1020	15	179° W, 35° N (4, 7)	5	This paper
3	Logan sills	R	1050	12	140° W, 47° N (12, 13)	8	This paper
4	Abitibi dikes*	N	1150†	10	134° W, 27° N (6, 7)	5	Larochelle (1966)
5	Mackenzie Igneous events	N	1200	154‡	171° W, 4° N	5	Fahrig and Jones (1969)

*Abitibi dikes are those referred to by Larochelle (1966) as having an ENEW direction. This follows the definition of Abitibi dikes used by Fahrig *et al.* (1965), namely those northeasterly dikes that are particularly well developed in the vicinity of Abitibi Lake. It may be argued that Larochelle's result, taken from only 5 dikes, is not reliable, and may contain a secular variation component: however, a previous study (Fahrig *et al.* 1965) of 14 sites gave a pole at 132° W, 32° N, which is not significantly different. A more comprehensive study of the Abitibi dike swarm will be published shortly.

†Biotite K–Ar age determinations (Fahrig and Wanless 1963) indicated an age of 1230 m.y. for the Abitibi intrusions; however, early G.S.C. K–Ar determinations tend to be consistently high and later K–Ar analyses have yielded ages of about 1100 m.y. As minor argon loss has probably occurred we consider that 1150 m.y. is the best estimate of the age of the Abitibi intrusions.

‡The Mackenzie pole position was calculated by Fahrig and Jones as the mean of 4 units. 107 of the sites came from one unit, named 'Muskox etc.', which included both Mackenzie dikes and Coppermine River lavas, and is very underweighted in the averaging process. Since all 4 poles lie close together a more fair weighting process would make an insignificant difference to the pole position, but the estimate of A_{95} is unduly conservative.

S = the number of sites from which mean directions were calculated. The site directions were obtained from the mean of at least 2, and commonly 5 or 6 separately oriented samples.

A_{95} is the half-angle of the cone of confidence at a probability P = 0.95, using the sites as units.

Polarity: M = Mixed, N = Normal, R = Reversed.

of the sills and nearly opposite to that of the other dikes. Two of these, D9 and D19, are wide, northwesterly striking dikes that can be traced for many miles (Fig. 2), and are probably feeders for the sills; in support of this view, both sites give initial wide scatters, which become closely grouped after cleaning in an alternating field of 400 Oe peak, typical of the sills.

The dikes at sites D3 and D6 strike northeast and we suggest a different origin for their magnetization, although in an area of cross-cutting dikes it is not always possible to be sure of the strike of the dike at the sampling site. They may belong to a set of dikes older than the sills, and may have been baked by them at the time of intrusion. The high blocking temperature (Fig. 3*a*) and coercivity spectra (Fig. 4*a*) are consistent with this possibility.

The mean site poles after cleaning (Fig. 5*a*) are widely scattered, and since they may have different histories no attempt has been made to mean them, although the mean direction from the reversely magnetized dikes (Table 4) is close to that for the sills.

Definition of the Logan Loop

More than 10 years ago DuBois discovered that paleomagnetic poles from late Proterozoic rocks of the Canadian Shield formed a great loop (DuBois 1962, p. 69), which (on the present geographic grid) swept north from the equator at 150° W longitude up into North America, thence west and south to the equator at 170° E longitude. At that time, however, the apex of the loop was poorly defined, and it was not clear how much of the movement could be attributed to secondary components in some of the rock formations. Also, although he knew in general the sequence of events, the absolute ages of the origin of most of his formations were not known, so that it was impossible to estimate the rate at which this movement took place.

Five relatively well-dated pole positions from late Proterozoic rocks of the Canadian Shield (Table 6) define critical points on the polar wandering loop. As the apex of the loop is anchored by poles derived from the Logan sills and Logan dikes, we propose that this major polar wandering movement in Neohelikian[2] time relative to the central Canadian Shield be called the Logan Loop. Other North American pole positions obtained by different workers (Table 7) fit this loop quite well. The rapid polar movement that caused the Logan Loop probably ceased about 1000 m.y. ago. After that time the pole appears to have stabilized at about 170°, 10° N,

[2]Neohelikian is the youngest (upper) division of the middle Proterozoic (Stockwell 1964).

TABLE 7. Corroborative pole positions

Symbol	Formation	Polarity	Age* (m.y.)	S	Pole position	A_{95}	References
A	Frieda and Nonesuch shales	N	—	13	170°E, 9°N (2, 4)	4	DuBois 1962
B	Portage Lake lavas	N	< 1115	14	177°W, 24°N (6, 10)	9	Vincenz 1968
C	Minnesota normal	N	—	131	168°W, 35°N	2	Beck 1970
D	North Shore volcanics normal	N	1115	14	173°W, 32°N	—	Palmer 1970
E	North Shore volcanics reversed	R	1115	54	160°W, 47°N	—	Palmer 1970 Age: Faure *et al.* 1969
F	Osler lavas	R	—	12	157°W, 49°N	9	Palmer 1970
G	Mugford	N	950		141°W, 45°N (8, 10)		Murthy and Deutch in press. Age: Wanless *et al.* 1966
H	Front Range	M	1410	5†	151°W, 8°S (4, 6)	4	Eggler and Larson 1968 Age: Ferris and Krueger 1964
J	St. Francois Mountains, Miss.	N		5‡	150°W, 5°N	—	Hays and Scharon 1966
K	St. Francois Mountains, Miss.	R	1300–1400	11	141°W, 1°S	5	Hsu *et al.* 1968
L	Michikamau Anorthosite	N	1400	6	145°W, 1°S	6	Murthy *et al.* 1968
M	Croker Island Complex	N	1475	19	143°W, 5°N (7, 11)	9	Palmer 1969

*Unlike Table 6, these formations are not necessarily in correct chronological sequence, as there is not adequate age control in many instances, and the reliability of the stated ages range widely.

†Each site is a separate petrological unit.

‡This is the mean of 5 units, of which the Stouts creek rhyolite was extensively sampled and 10 or more samples were taken from the remaining units.

S = the number of sites, such as a quarry, single rock outcrop, or lava flow, that were sampled. However, the method of sampling varied widely – some sites are represented by a single sample of unknown reliability, while from others the direction was obtained from the mean of a group of independently oriented samples, so that the reliability of the direction could be assessed.

A_{95} = the half-angle of the cone of confidence at a probability $P = 0.95$, using the sites as units.

for as long as 300 m.y. The time of commencement of the movement that outlines the loop is uncertain, but 1200 m.y. is the older limit. Prior to this time the pole was stabilized for a period of about 300 m.y. at a position of 160° W, 0°. These 'quasi-static intervals' in Neohelikian and Paleohelikian time are comparable with those described by Briden (1967) for Phanerozoic time. (e.g. Irving *et al.* 1963; Gough *et al.* 1964). They are longer by a factor of nearly two than the longest recognized during the Phanerozoic.

If the polar shift of 31° between the times of intrusion of the normal dikes and the Logan sills, and the 20° shift between the intrusion times of the Logan sills and the Abitibi dikes was spread evenly over a period of 130 m.y., an annual movement of 4.3 cm/year is indicated. Tanton (1931), however, considered on geological grounds that the normally magnetized northeast-striking dikes and the Logan sills were of similar age, and the available K–Ar ages of Table 8 support this view.

Assuming 10 m.y. between these two intrusive events, a uniform rate of movement gives a rate of 34 cm/year. This speed is comparable with the rapid polar wandering suggested by Irving and Robertson (1969) for episodes such as the Paterson transition in the Carboniferous Period.

Until the ages of intrusion of the normally magnetized dikes and the Logan sills are more precisely determined, the rate of polar wandering movement during and between the times of these igneous events must be considered to have a possible range from slow ocean spreading, such as that about the mid-Atlantic Ridge today (Heirtzler *et al.* 1968) (about 1 cm/year), to fast polar wandering (up to 30 cm/year). The Logan Loop does show in general, however, that rates of polar movement of the same order of magnitude as those in the Phanerozoic also took place

TABLE 8. Radiogenic ages of sills and dikes

Sample No.	Method	Location	Age $\times 10^{-6}$ years	Reference
Sills				
PA-3	K–Ar Whole-rock	48°14′N, 89°59′W	1030 ± 40	T. J. Ulrych in Palmer (1970)
PA-16	K–Ar Whole-rock	48°16′N, 89°35′W	1060 ± 40	" " " " "
NP-6	K–Ar Whole-rock	49°26′N, 88°07′W	865 ± 60	N. J. Snelling in Palmer (1970)
NP-138	K–Ar Whole-rock	49°19′N, 88°07′W	730 ± 50	" " " " "
67FR478	K–Ar Chilled margin	48°18′N, 89°40′W	924 ± 90	G.S.C. (J. M. Franklin unpubl.)
67FR476	K–Ar Whole-rock	48°59′N, 88°20′W	1000 ± 140	G.S.C. " " "
67FR475	K–Ar Chilled margin	48°59′N, 88°20′W	963 ± 134	G.S.C. " " "
67FR472	K–Ar Chilled margin	49°30′N, 88°03′W	837 ± 115	G.S.C. " " "
Dikes				
67FR482	K–Ar Whole-rock	48°04′N, 89°31′W	1045 ± 40	G.S.C. (J. M. Franklin unpubl.)
67FR479B	K–Ar Biotite	48°04′N, 89°35′W	993 ± 35	G.S.C. " " "
FA-11-69	K–Ar Whole-rock	48°20′N, 88°49′W	998 ± 40	G.S.C. (W. F. Fahrig unpubl.)
G.S.C.61-138	K–Ar Biotite	48°01′N, 89°30′W	1000 ± 60	Lowden *et al.* (1963)
G.S.C.64-101	K–Ar Biotite	48°02′N, 89°30′W	1055 ± 40	Wanless *et al.* (1966)

in Neohelikian time. This eliminates the possibility that all polar wandering curves can be explained by a single drawnout episode of continental drift resulting from the breakup of the super-continent of Pangaea, first into the two parts of Gondwanaland and Laurasia, and then into the seven continents of today. The Logan Loop indicates that continental drift and/or polar wandering took place in Neohelikian time unless the earth's field had quite a different configuration from that of today.

The rapid polar shift indicated by the Logan Loop occupied the same period of time as the Grenvillian orogeny. If orogenies generally reflect ocean-floor spreading and continental drift (Irving and Robertson 1969), then a period of rapid apparent polar wandering, expressing continental drift, would be expected to coincide with orogenic episodes. It is suggested then, that the Logan Loop and the Grenvillian orogeny are genetically related.

The Grenville orogenic belt has been traced through western Scotland and Scandinavia (DuToit 1937), and it has been suggested that at that time, before the birth of the Atlantic, the Baltic and Canadian Shields were united. If so they should have moved as a single unit, and a paleomagnetic study of rocks of this age from the Baltic Shield should yield a polar loop similar to the Logan Loop (Fahrig *et al.* 1971).

Paleolatitudes and the Geological Test

We will here assume the axial dipole field hypothesis, so that when short term irregularities have been averaged out, the earth's magnetic field is symmetrical about the axis of rotation, the geographic pole.

Each of the 5 key pole positions (Table 6) is the mean from a large number of sites thought to span enough time to average out short term irregularities such as secular variation. They may therefore accurately represent the geographic pole at their time of formation, and are used in Fig. 6 as the basis for paleolatitude maps for each of these times.

During Franklin time the equator passed meridionally through Winnipeg, and the Mackenzie time equator passed east of Hudson Bay and the Great Lakes. Hence, for the two quasi-static intervals on either side of Logan Loop time the central Canadian Shield was in the tropics. Unfortunately in times as remote as those the present is a very imperfect key to the past. Apart from stromatolites, fossils diagnostic of latitude are absent, and only environmental features of sediments may be available as indicators of paleolatitude. Evaporites and chemically precipitated limestones are to be expected. Evaporites (anhydrite) in the Shaler Group are probably closely related in time to the Franklin event, and the chemical precipitates of the Gunflint Formation (Taconite) probably relate to the quasi-static interval terminated by the Mackenzie event. Stromatolite horizons occur in the Sibley Series, dated at 1300 m.y. (see discussion under the section headed "Geological Setting, Age, Sampling, and Measurement").

The situation at the time of intrusion of the

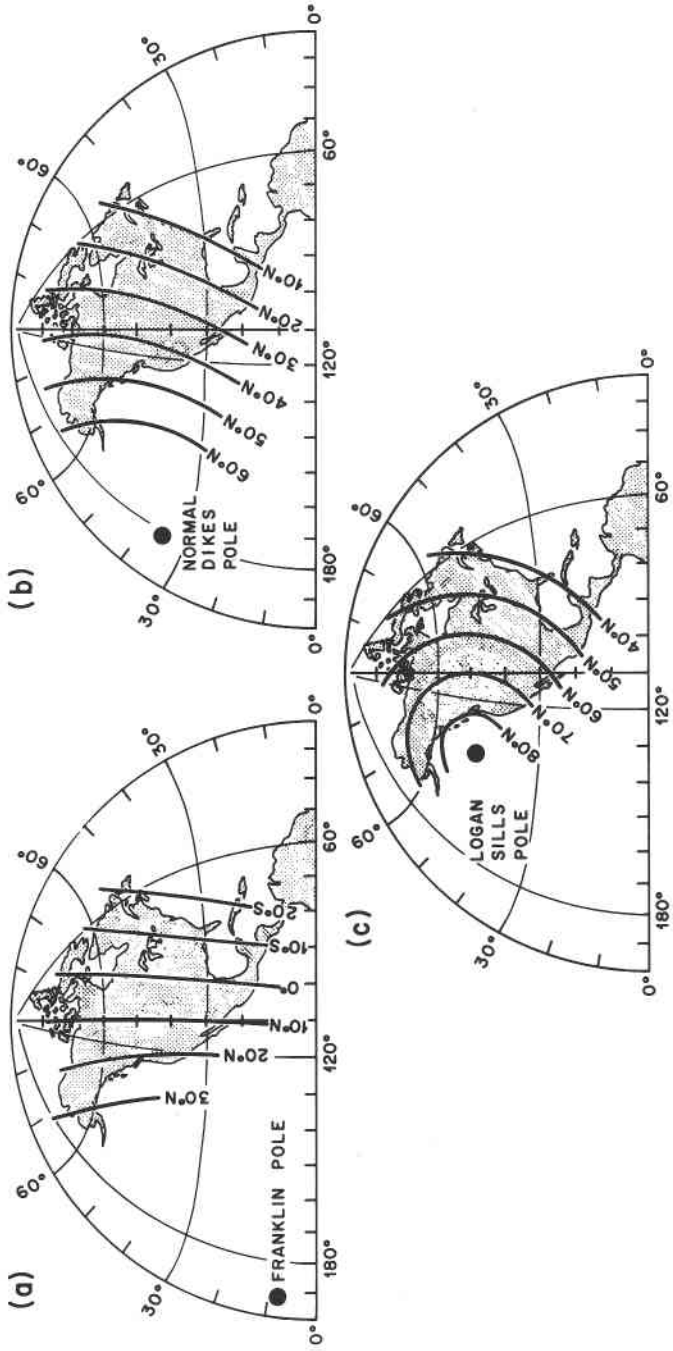


FIG. 6. Paleolatitude maps showing the lines of paleolatitude in North America relevant to (*a*) The Franklin pole, (*b*) the normal dike pole, (*c*) the Logan sills pole, (*d*) the Abitibi dike pole, and (*e*) the Mackenzie pole.

Logan sills and Abitibi dikes is quite different. It is evident in Figs. 6*c* and *d* that the paleolatitudes calculated from these poles place the central Canadian Shield in high latitudes, where glacial or periglacial conditions become a possibility. Unfortunately, the sedimentary record for Neohelikian times is so fragmentary, and our knowledge of climatic zoning so scanty (the thermal regime may well have been greatly modified by different land distribution and possibly differences in heat intake from the sun caused either by differences in the sun's radiation or the absorption of the atmosphere), that a wide range of possibilities exists as to the climate to be expected at any latitude and the sediments that would form. However, any evidence of Canadian Shield glacial deposits similar in age to that of the Logan sills or the Abitibi dikes would strongly support inferences based on the high paleolatitudes indicated by the paleomagnetism of these rocks.

The greatest single obstruction to clarifying and testing the validity of the Logan Loop, as shown in Fig. 7, is the difficulty of obtaining absolute and relative ages within the Helikian Era of sufficient accuracy to be able to identify contemporaneous rocks from different parts of the shield. Perhaps the next advance will have to await a better-defined Helikian time scale, based both on radiogenic and paleomagnetic criteria.

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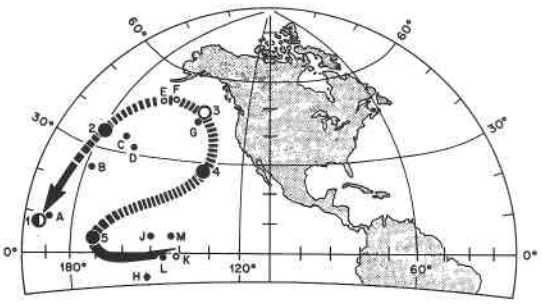


FIG. 7. An equal area projection of the western hemisphere showing the Logan Loop on the polar wandering curve relative to North America in Hadrynian and Helikian time. The numbers refer to the pole positions of Table 6 and the letters to those of Table 7.

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NOTE ADDED IN PROOF (SEPTEMBER 1971):

A recent interpretation of the Precambrian polar-wandering curve for North America was published by Spall (1971) after this paper was written. The most westerly (younger) part of his curve (Fig. 1) corresponds in time to our Fig. 7: the greater depth of our loop is the result of the inclusion of pole positions from the Logan sills, the North Shore volcanics, and the Osler lavas (3, E, F), three formations that yield reversed magnetizations.

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